No.	Status	Date
HIS-2007-0388-DS-2-RV-102	Issued	2007-09-18

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RV-102 located within the ATU-100. The scope of work is limited to the thickness data collection for this specific line as part of the broader survey detailed in HIS-2007-0388.

Kevin Doucette, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the technical requirements of the project and the site-specific quality standards.

Technical Basis and Procedures

The thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 0 and HIS-UT-002 Rev 1. These documents establish the site NDE procedure for the ATU-100 and the specific ultrasonic thickness measurement methodology submitted by Halward Inspection Services Ltd.

The inspection methodology and instrument calibration were conducted in accordance with ASTM E797. All readings were taken using the ATU-100 approved equipment to ensure consistency with previous in-service NDE records.

Component Design Data

The subject line, 2-RV-102, is a regenerator overhead line constructed from A106 Gr.B. The nominal outer diameter is 60.3 mm with a nominal wall thickness of 5.54 mm.

The design parameters for 2-RV-102 are established at a maximum allowable pressure of 345 kPa(g) and a design temperature of 121 °C. The minimum allowable thickness (t-min) for this component is 2.79 mm.

Evaluation Criteria

Acceptance is based on the retirement thickness, which is defined as the t-min of 2.79 mm. Any thickness reading recorded below this value is subject to an NCR.

Corrosion rates are determined by comparing the long-term rate since the first reading against the short-term rate of the latest pair of readings taken at least 3 years apart. The remaining life is calculated as (t - t-min) / rate.

Inspection Interval

The next thickness reading interval is determined by half the remaining life, provided it does not exceed the class maximum. Line 2-RV-102 is categorized as Class 1 due to its service as a sour gas regenerator overhead.

For Class 1 components, the maximum interval for thickness measurements is 5 years and the maximum interval for external visual inspection is 5 years.

Thickness readings, measured 2007-09-18

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2006-02-24	6.98	6.95	0.056	0.047	>50	2012-09-18	
3-RV-101-02	7.62	3.56	2006-02-24	6.93	6.89	0.059	0.064	>50	2012-09-18	
3-RV-101-03	7.62	3.56	2006-02-24	6.57	6.56	0.083	0.090	33.3	2012-09-18	
3-RV-101-04	7.62	3.56	2006-02-24	6.65	6.66	0.081	0.079	38.4	2012-09-18	
3-RV-101-05	7.62	3.56	2006-02-24	7.01	6.87	0.069	0.058	47.7	2012-09-18	
3-RV-101-06	7.62	3.56	2006-02-24	6.74	6.71	0.076	0.075	41.3	2012-09-18	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2006-02-24	4.85	4.83	0.061	0.070	29.0	2012-09-18	
2-RV-102-02	5.54	2.79	2006-02-24	4.84	4.67	0.080	0.069	23.6	2012-09-18	
2-RV-102-03	5.54	2.79	2006-02-24	5.14	5.01	0.041	0.035	>50	2012-09-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2006-02-24	5.07	5.12	0.040	0.032	>50	2012-09-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2007-09-18	_____